How a photon tells oxygen from nitrogen

What physically separates the species, and how well is that separation known?

Notebook companion: Unit-04-cross-sections.ipynb

1 The question

An occultation measures attenuation and nothing else. The sensor watches the Sun set behind the Earth's limb. As the line of sight descends into the atmosphere, more gas lies along the path and the signal falls. What each channel returns is one number for each tangent altitude: the fraction of the sunlight that survived the passage.

That number does not say which gas absorbed the missing light. If atomic oxygen and molecular nitrogen absorbed sunlight identically, a set of channels would tell us only how much gas lay along each path. We would measure total mass density and learn nothing about composition.

They do not absorb identically, and this unit is about the difference. There are four questions, in order: what causes the difference; at which wavelengths it appears; how much of it survives once a real filter has averaged over a band of wavelengths; and how accurately it is known. The last question is the one that limits what we are allowed to claim.

Two later results depend on the answers. Unit 10 chooses the channel set according to how well each filter separates the two species. Unit 8 sets the accuracy floor of the whole measurement, and that floor is fixed by how well the cross sections are known. Collecting more photons does not move it.

2 Physical picture

A photon is removed from the beam when a particle absorbs it. The probability of that happening, per unit column of gas along the path, is the photoabsorption cross section $\sigma(\lambda)$. It has units of area and is quoted in megabarns, where $1\,\mathrm{Mb} = 10^{-18}\,\mathrm{cm}^2$. It can be read as the effective target area that each particle presents to a photon of that wavelength. It depends on the species and the wavelength only — not on the atmosphere, the geometry or the instrument.

How large σ is at a given wavelength depends on which absorption processes the photon carries enough energy to trigger. Three of them matter here.

Ionization occurs when the photon carries more energy than it takes to remove an electron. The threshold is 13.6 eV for atomic oxygen, which corresponds to a wavelength of 91.1 nm; 15.58 eV, or 79.6 nm, for N_2; and 12.07 eV, or 102.6 nm, for O_2. Below each of those wavelengths the species can be ionized; above it, it cannot.

Dissociation occurs when a molecule absorbs a photon and breaks apart. Only molecules can do this. An atom has no bond to break, and this asymmetry is the basis of everything that follows.

Predissociation is a second molecular route to the same outcome. The photon lifts the molecule

into a bound excited state, that state couples to another state in which the molecule is unstable, and the molecule comes apart. The absorption is strong, but it occurs in discrete bands at particular wavelengths rather than smoothly across the spectrum. Between 80 and 100 nm, N_2 has roughly a hundred such bands, most of which predissociate, and the upper levels of many are strongly perturbed [11].

Now put the three together. Above 79.6 nm, N_2 can no longer be ionized, but dissociation and predissociation are still available to it, so it keeps absorbing strongly out to about 99 nm. Atomic oxygen above 91.1 nm can no longer be ionized either, and being an atom it has neither of the other two routes. No absorption process remains open to it, so it becomes transparent.

Between 91.1 and 99 nm, therefore, nitrogen absorbs and oxygen does not. This is the one part of the band where the two species behave completely differently, and section 4b is about how hard it is to exploit.

Above 99 nm neither species absorbs. Molecular oxygen begins absorbing again near 130 nm, in the Schumann–Runge continuum, and reaches its peak near 142 nm. It is the only absorber left above 115 nm, which is why the far-ultraviolet channels see O_2 alone.

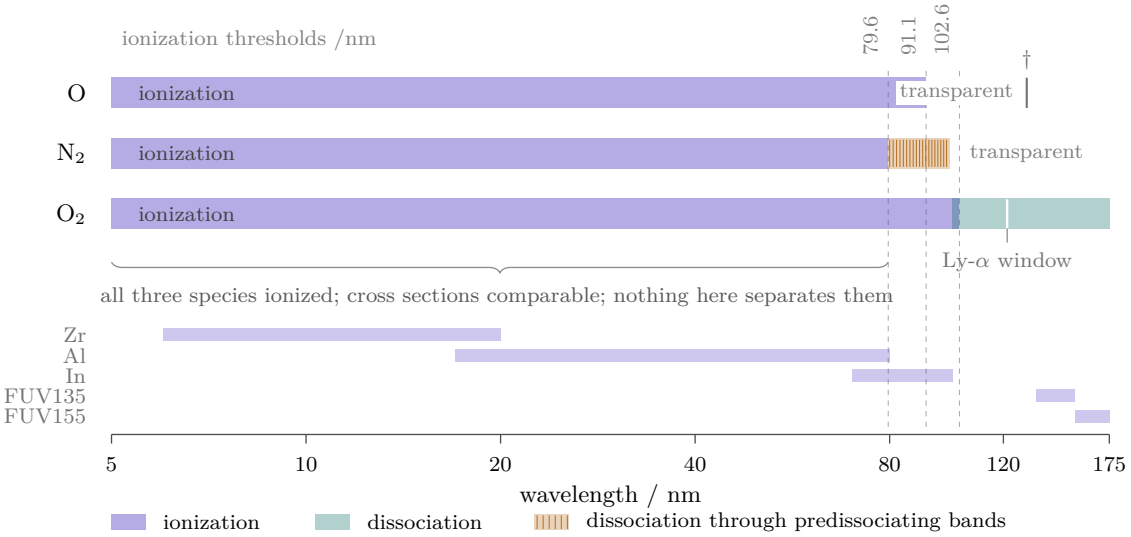


Figure 1: Which absorption processes are open to each species, against wavelength. Colour here means process, not species; Figure 2 colours by species. Ionization remains available at every wavelength shorter than the axis shows, so the three ionization bars all begin at the left edge. The figure makes both halves of the separation problem visible at once. On the left, all three species are ionized and nothing distinguishes them, which is why the zirconium and aluminium channels separate O from N_2 as weakly as section 4a finds. On the right, N_2 keeps absorbing to about 99 nm through dissociation and its roughly one hundred predissociating bands, while O, having no route open but ionization, becomes transparent above 91.1 nm. O_2 absorbs again from about 100 nm, and the strong Schumann–Runge continuum occupies 130–175 nm; the white line at 121.6 nm marks the narrow window in which O_2 is nearly transparent, drawn one line wide because on this scale its true 1 nm width would be invisible. † At 130.4 nm, atomic oxygen removes light from the beam by resonant scattering. That is not photoabsorption, it is absent from the cross-section compilation, and section 7 explains what it costs us. The logarithmic axis compresses the far ultraviolet: 100–175 nm occupies the last 16 % of its width.

3 Derivation: from a cross section to a channel number

A beam of a single wavelength obeys Beer–Lambert exactly. Our channels pass a band of wavelengths, and that changes the arithmetic in a way that affects the numbers we report.

The detector cannot distinguish photons of different wavelength within one channel; it counts them all together. What the channel measures is therefore the total transmitted flux divided by the total incident flux, which is the flux-weighted average of the single-wavelength transmission. Writing F_i for the solar photon flux in wavelength bin i, r_i for the filter response there, σ_{si} for the cross section of species s in that bin, and N_s for the slant column of that species,

$$T_{\mathrm{ch}} \;=\; \frac{\sum_i F_i r_i \exp(-\sum_s \sigma_{si} N_s)}{\sum_i F_i r_i}\,, \qquad \tau_{\mathrm{ch}} \;=\; -\ln T_{\mathrm{ch}}\,.$$

This is what `occ.tau` computes. It does the arithmetic in log space so that a fully opaque channel returns a large finite number rather than underflowing to zero.

The average of an exponential is not the exponential of an average. There is therefore no single cross section σ for which $\tau_{\mathrm{ch}} = \sigma N$ holds at every column. One does exist when the absorption is weak. If every $\sigma_{si} N_s$ is small, expanding the exponential to first order leaves a plain flux-weighted mean,

$$\sigma_s^{\mathrm{eff}} \;=\; \frac{\sum_i F_i r_i \sigma_{si}}{\sum_i F_i r_i}\,.$$

That is the natural meaning of the phrase "the cross section of a channel", and it is what every table in this unit reports. It is also what a reader will assume we mean when we quote such a number, so it is worth being explicit about where it stops being accurate.

It stops being accurate once the absorption is strong, and it always fails in the same direction. As the column grows, the wavelengths inside the passband that absorb most strongly go opaque first, and almost no light from those wavelengths reaches the detector. The light that does arrive is then dominated by the wavelengths that absorb least. The channel's apparent cross section therefore falls as the column rises. Astronomers call the resulting relationship between column and measured absorption a curve of growth.

Section C of the notebook measures the size of this effect for the aluminium channel. At an optical depth of 0.017 the weak-absorption formula is accurate to 0.3 %. At 0.51 it overestimates the optical depth by 8 %, at 1.43 by 22 %, and at 3.49 by 58 %.

Two consequences follow. The first is that every effective cross section in this unit is a weak-absorption value. They are for reasoning about which channel sees which species. They must not be used to convert a measured optical depth into a column density; the forward model does that correctly, by integrating over the band. The second is that this is one of the two reasons the usable optical-depth window has an upper limit at $\tau = 4$. Above it, the channel is reporting mainly on its least-absorbing wavelengths rather than on the gas. The lower limit of the window is set by photon noise and belongs to Unit 6.

4 Worked numbers

All the numbers below come from the compilation in `channel-plan/data/`, which tabulates σ for O, N_2 and O_2 together with a reference solar spectrum, on a shared grid of 123 wavelength bins

running from 0.05 to 175 nm. The bins are 1 nm wide through most of the extreme ultraviolet and widen to 5 nm above 121 nm. Every filter band is treated as an ideal top hat: full transmission inside its stated limits, none outside.

4a. The channels we have

Channel	Band /nm	$\sigma_{\mathrm{O}}^{\mathrm{eff}}$	$\sigma_{\mathrm{N_2}}^{\mathrm{eff}}$	$\sigma_{\mathrm{O_2}}^{\mathrm{eff}}$	$\mathrm{N_2/O}$	Band flux
Zirconium	6–20	3.14	4.10	6.31	1.3	2.29×10^{9}
Aluminium	17–80	8.79	17.42	19.57	2.0	2.11×10^{10}
Indium	70–100	2.26	12.60	16.05	5.6	2.10×10^{10}
FUV	135–155	0	0	11.83	—	1.09×10^{11}
FUV	155–175	0	0	1.76	—	7.13×10^{11}

Cross sections in Mb; band flux in photons $\mathrm{cm^{-2}\,s^{-1}}$. Weak-absorption flux-weighted averages over ideal top-hat bands.

The $\mathrm{N_2/O}$ column is the one to read. We call that ratio the channel's *contrast*: it says how differently the two species appear to that channel. A channel whose contrast is 1 absorbs equally from both and cannot tell them apart at all.

Zirconium and aluminium are the two channels the baseline design depends on, and their contrasts are 1.3 and 2.0. Both are low. More importantly, they are close to each other. Recovering two densities from two channels means solving two equations whose coefficients are those contrasts, and when the two sets of coefficients are nearly proportional, a small error in the measured optical depths turns into a large error in the recovered densities. Indium, at 5.6, is the only channel anywhere in the band whose contrast differs strongly from the other two.

This is a statement about cross sections alone. No atmosphere, no retrieval algorithm and no noise model has entered it. It is where the conditioning argument of Unit 10 and the degeneracy measured in Unit 8 originate, and neither of those decisions is taken here.

4b. The window between 91.1 and 100 nm

Between 91.1 and 100 nm, nitrogen absorbs and oxygen does not. The team brief quotes the ratio of their cross sections there as 136. Our calculation gives 136.5 for a filter that passes everything between 91.1 and 100 nm and nothing outside it.

That ratio depends almost entirely on where the lower edge of the filter falls. Oxygen absorbs strongly just below 91.1 nm and not at all above it, so the ratio is in effect a measure of how much of the passband lies below 91.1 nm. A filter beginning exactly at 91.1 nm admits no oxygen absorption and the ratio is large. A filter beginning one nanometre lower admits a slice of strong oxygen absorption, and the ratio falls by more than a factor of twelve:

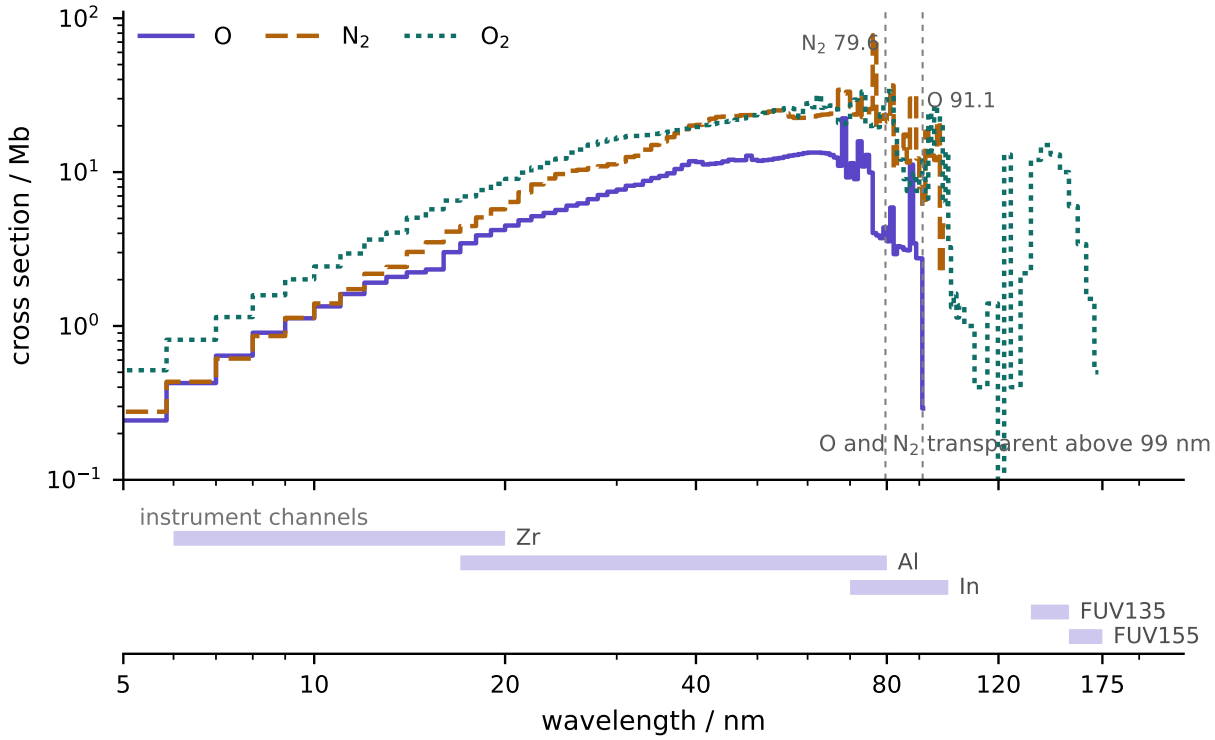


Figure 2: Photoabsorption cross sections for the three major species, with the five instrument channels beneath them on the same wavelength axis. Both thresholds that matter are visible directly in the tabulated data: N_2 keeps absorbing strongly above its ionization threshold at 79.6 nm, while O falls to zero above 91.1 nm. Above 99 nm only O_2 absorbs. Generated by section D of the notebook.

Band /nm	$\sigma_{\mathrm{O}}^{\mathrm{eff}}$	$\sigma_{\mathrm{N_2}}^{\mathrm{eff}}$	$\sigma_{\mathrm{O_2}}^{\mathrm{eff}}$	$\mathbf{N_2/O}$
90.0–100	0.4473	4.89	16.81	10.9
90.5–100	0.2580	4.63	17.40	17.9
91.1–100	0.0316	4.31	18.16	136.5
91.5–100	0.0184	4.22	18.48	229.3
92.0–100	0	4.11	18.93	∞

Moving the edge in the other direction, to 92.0 nm, makes the ratio infinite, because the oxygen cross section is exactly zero there in this compilation. Nothing about the molecules has changed anywhere in this table. What changes is how much of the filter's passband lies on the absorbing side of the oxygen threshold.

Real filters do not have sharp edges. A filter that leaked even a few per cent of its transmission below 91.1 nm would land somewhere on the steep part of this table, and we cannot say where until a vendor supplies a measured transmission curve. Two things follow, and we act on both. No channel in the plan is justified by this window. And the vendor request for information asks filter houses for measured curves, including the shape of the edges and the out-of-band rejection, rather than for a nominal passband — because this table is what a nominal passband is worth.

One further point about the same window. The O_2 cross section there is about 18 Mb, which is

larger than the N_2 value of roughly 4 Mb. Oxygen molecules absorb this light more strongly than nitrogen molecules do. The window is therefore a clean nitrogen channel only at altitudes high enough that O_2 has become a minor constituent.

4c. Lyman-alpha

Lyman-α, at 121.6 nm, is a problem for any far-ultraviolet channel whose passband reaches down to it. Two facts combine. The line is extremely bright: in this compilation its wavelength bin carries 3.79×10^{11} photons cm^{-2} s^{-1}, which is 28.8 % of all the photons the Sun emits between 0.05 and 175 nm. And molecular oxygen barely absorbs at that wavelength — 0.01 Mb, against about 12 Mb in the Schumann–Runge continuum. Solar Lyman-α reaches the mesosphere for exactly this reason.

A channel that admits the line therefore collects most of its light at a wavelength where the atmosphere is nearly transparent. Those photons arrive whether or not there is any oxygen along the path, and they dilute the absorption the channel exists to measure. Widening the 135–155 nm band down to 121 nm cuts its effective oxygen cross section from 11.83 to 2.74 Mb, a factor of 4.31. The channel would then need more than four times as much oxygen along the path to register the same absorption, and the altitude at which it became useful would drop accordingly.

Extending the band only as far as 130 nm costs much less: the effective cross section falls from 11.83 to 9.98 Mb, a factor of 1.18. The reason to stop short of 130 nm is a different one, and section 7 explains why this calculation cannot demonstrate it.

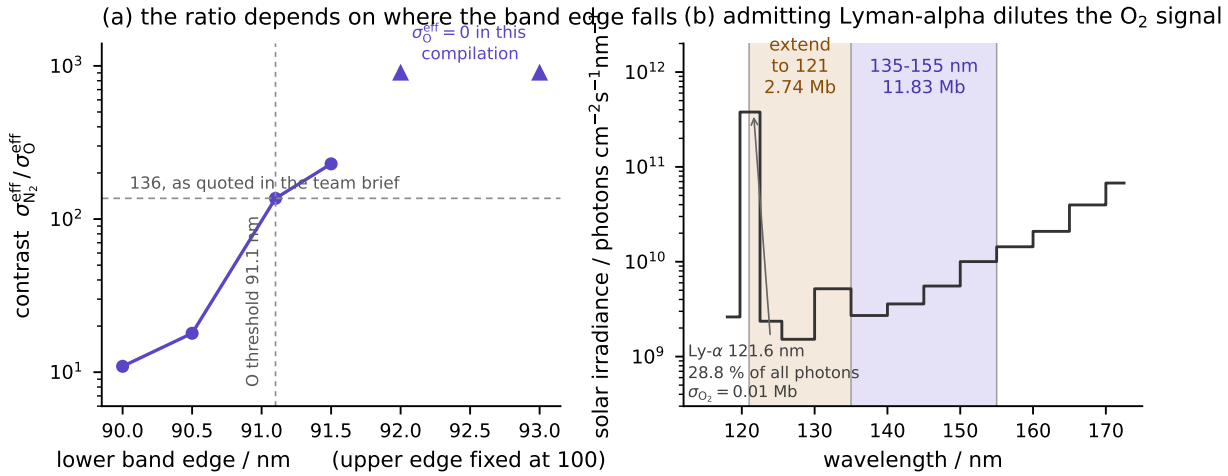


Figure 3: (a) The nitrogen-to-oxygen cross-section ratio for a band ending at 100 nm, plotted against where the lower edge of the band falls. The dashed line marks the value quoted in the team brief. Above 92 nm the oxygen cross section is identically zero in this compilation and the ratio is infinite. (b) The solar spectrum between 112 and 175 nm, with the two candidate far-ultraviolet bands. Extending the band down to 121 nm admits Lyman-α and cuts the effective O_2 cross section by a factor of 4.3. Generated by sections E and F of the notebook.

5 What the computation shows

Three results, in the order in which they are used again.

The contrast between two species belongs to the band, not to the molecules. The molecular thresholds are sharp and well established. The numbers an instrument actually experiences are averages over a passband, weighted by how much sunlight arrives at each wavelength inside it, and they depend on where that passband begins and ends as strongly as they depend on the physics. A contrast quoted without its band edges is not a specification.

Zirconium and aluminium separate the species only weakly. Zirconium's contrast is 1.3 and aluminium's is 2.0. Because those two numbers are close, small measurement errors produce large errors in the recovered densities, and that is true before any noise model or retrieval algorithm has been introduced. Unit 8 measures the consequence and Unit 10 decides what to do about it.

A single bright solar line can determine what a channel measures. Lyman-α is the clearest example in the series of why the forward model integrates over a measured solar spectrum rather than evaluating a cross section at a channel's nominal centre wavelength.

6 What we decided

The far-ultraviolet passbands exclude 121.6 nm. The justification is the factor of 4.31 in section 4c, not a rule of thumb.

We do not claim the window between 91.1 and 100 nm as a design feature of the instrument, and no channel in the plan depends on it. Section 4b is why.

The vendor request for information asks filter houses for measured transmission curves rather than nominal passbands. This unit is the argument behind that question. In the worst case in section 4b, the difference between a nominal filter edge and a real one is a factor of twelve in the very quantity the channel exists to provide.

Every effective cross section quoted in this unit is a weak-absorption value. The forward model does not use them; it integrates over the band. They are here to support reasoning about channel design, and they should not be used to turn a measured optical depth into a column density.

7 What we do not claim

Not that σ is known to the precision of the tabulated digits. The data files carry four decimal places. The measurements underneath them do not support that many. The standard compilation for this band is Fennelly and Torr [2], and neither it nor the principal underlying measurements of atomic oxygen [1, 7] states an explicit percentage uncertainty that we were able to find. The uncertainties that do appear attached to these cross sections come from the people who use them rather than from the people who measured them. Sewell et al. [8], for instance, tabulate σ_{O} as 3.37 ± 0.11 at 17.1 nm and 7.67 ± 0.28 at 30.4 nm, in units of $10^{-18}\,\mathrm{cm}^2$; those are fractional uncertainties of two to four per cent, which is our arithmetic on their numbers rather than a figure they quote. For stable species measured directly, Heays et al. [4] give the general range as 10 to 20 per cent.

Not that a grid of 123 bins resolves the structure it is averaging over. Between 80 and 100 nm the nitrogen spectrum is a dense system of narrow bands [11]. Heays et al. warn that averaging over line structure that has not been resolved can underestimate an integrated cross section "potentially by more than an order of magnitude." Solomon and Qian [9] report that

binning errors in their scheme stay below about 5 %, but their scheme uses 22 intervals and ours uses 123, so their figure does not transfer to our grid; they add that the uncertainties in the cross sections themselves are "far greater than those introduced by our parameterization." Two further results indicate how large the effect of resolution can be. Meier et al. [5] find errors of up to a factor of three in lower-thermospheric oxygen photoionization rates computed at 1 nm resolution. Soto et al. [10] report that the choice of resolution alone changes O^+ production rates by more than 500 %. Both concern production rates rather than occultation columns, so neither number transfers directly to our measurement, but together they establish that the resolution of the grid is not a rounding question.

Not that resonance scattering is represented anywhere in this unit. The compilation contains photoabsorption cross sections only. A far-ultraviolet band must also exclude the atomic oxygen resonance triplet at 130.4 nm, and the reason is that thermospheric atomic oxygen scatters light at that wavelength resonantly, removing it from the beam by a process that is not photoabsorption and is not in these data files. Section 4c finds that extending the band to 130 nm costs only 18 % of the effective cross section. That figure is an underestimate of the true cost, for exactly this reason. The exclusion of 130.4 nm is argued physically; it is not computed here.

Not anything about how σ varies with temperature. Every value used here is a room-temperature value. In the Schumann–Runge continuum, measurements at 78 K have come in 20 % below the 295 K values, while theory predicts a difference of only a few per cent [6]; and the historical spread among independent determinations of the same continuum was itself 10 % or worse [3]. The thermosphere is not at 295 K.

Not that the bands used here are the real filter response. Every band in this unit is an ideal top hat. This is the first of the project's four standing caveats, and it is the one this unit is best placed to quantify.

It is worth stating plainly how much the paragraphs above concede, because this is the accuracy floor that Unit 8 starts from. In the closest existing measurement to ours, Sewell et al. [8] report random uncertainties at 250 km of ±8 % for O and $-17/+16$ % for N_2, and they identify N_2 as the largest single term in their error budget. They also found that adjusting the Fennelly and Torr cross sections by at most about 5 % reduced the scatter between their retrievals and independent measurements by roughly 25 % for O and 45 % for N_2 — which is a direct indication that the tabulated cross sections are wrong by something of that order. Thiemann et al. [12] carry an explicit cross-section term in their systematic error budget and report a total systematic error below 13 % and a random error below 6 %. None of these figures is a limit imposed by hardware. They are limits imposed by how well the community knows σ, and a better detector does not improve them.

8 Exercises

E4.1 Recompute the table in section 4a with the aluminium band narrowed to 17–60 nm. Report both the change in contrast and the change in the total flux the band collects. One of those two governs how well the channel separates the species; the other governs the range of altitudes over which the channel's optical depth falls inside the usable window. Say which is which, and state what you would need to know about the atmosphere before you could decide whether the narrower band is worth adopting.

E4.2 Change the 135–155 nm band to 121–155 nm and recompute the effective cross section for O_2. Then, for each version of the band, find the O_2 column at which the channel reaches an optical depth of 1. Use the difference between the two answers to write a short statement about what is lost when a filter is specified only by its nominal centre wavelength.

E4.3 The contrasts in section 4a are averages weighted by a single reference solar spectrum. Solar extreme-ultraviolet irradiance varies by more than a factor of two over the solar cycle, and it does not vary by the same amount at every wavelength — the shortest wavelengths vary most. Propose a test of whether the effective cross sections are stable enough across the cycle to be treated as constants in the forward model. Say what data the test would need, and what you would do if it failed for one channel and passed for the others.

References

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Every quantitative result in this unit is provisional in the four ways that apply to the whole project: filter bands are idealized top-hats rather than measured transmission curves; the truth atmosphere is NRLMSIS 2.1; helium is omitted above 400 km; and no horizontal structure along the ray is modelled. Analysis code is in `channel-plan/`.